



Poisson Distribution



Poisson random variable

Another discrete random variable

Provides a good model for data that represent the number of occurrences of a specified event **in a given unit of time or space.**



Example

- The number of calls received by a switchboard during a given period of time
- The number of bacteria per small volume of fluid
- The number of customer arrivals at a checkout counter during a given minute
- The number of machine breakdowns during a given day
- The number of traffic accidents at a given intersection during a given time period



In each example, x represents the number of events that occur in a period of time or space during which an average of μ such events can be expected to occur.

The only assumptions needed when one uses the Poisson distribution to model experiments such as these are that the counts or events occur **randomly and independently** of one another.



Let μ be the average number of times that an event occurs in a certain period of time or space. The probability of k occurrences of this event is

$$P(x = k) = \frac{\mu^k e^{-\mu}}{k!}$$



for values of $k = 0, 1, 2, 3, \dots$. The mean and standard deviation of the Poisson random variable x are

$$\begin{array}{ll} \text{Mean:} & \mu \\ \text{Standard deviation:} & \sigma = \sqrt{\mu} \end{array}$$



The average number of traffic accidents on a certain section of highway is two per week. Assume that the number of accidents follows a Poisson distribution with $\mu = 2$.

- Find the probability of no accidents on this section of highway during a 1-week period.
- Find the probability of at most three accidents on this section of highway during a 2-week period.



The Poisson probability distribution provides a simple, easy-to-compute, and accurate approximation to binomial probabilities when n is large and $\mu = np$ is small, preferably with $\mu < 7$